Moamen Elsharkawy

AI Automation Engineer | Fractional Chief AI Officer

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LinkedIn | GitHub | Portfolio

Professional summary

AI Automation Engineer building AI agents, n8n workflows and API-connected business systems with Python and TypeScript. Hands-on fractional Chief AI Officer who implemented tailored AI Operating Systems for two Cairo real-estate clients, alongside my own AIOS; both client systems are used internally. Experience includes CRM integration, tested lead pipelines, RAG workflows and paid transcript-translation delivery.

Technical skills

Automation: n8n, REST APIs, webhooks, CRM integration, retries, idempotency and audit trails

Applied AI: LLM APIs, tool calling, RAG, structured outputs, evaluation, MCP and human approvals

Development: Python, TypeScript, Node.js, FastAPI, PostgreSQL, React, Angular, Git and Docker

AI delivery: process discovery, roadmap prioritization, architecture, testing and documentation

Professional experience

Chief AI Officer (part-time) | Home Locator Aug 2026 - Present | Cairo, Egypt

- Define AI and technology priorities with the owner; translate operational needs into a roadmap and implement the selected workflows and integrations.
- Build new components and improve the existing CRM and website for customer management and sales-agent workflows used internally; own implementation and troubleshooting.

Chief AI Officer (part-time) | Real Experts Jul 2026 - Present | Cairo, Egypt

- Built the original Egyptian-Arabic AI Operating System and CRM from scratch; both are used internally to support brokerage operations.
- Define the AI roadmap with the owner and turn brokerage processes into operating rules, structured data and connected workflows; own architecture and implementation.

Additional experience

AI Automation Course Developer and Instructor, contract | EYouth Aug 2026 - Present

- Develop applied automation material and demos. Recorded Levels 1 and 2, both awaiting publication; filming Level 3.

AI Automation Engineer and Instructor | Sparkling for Business LLC May 2026 - Sep 2026 | Cohort-based

- Built 13 validated instructional workflows, led office hours and supported applied student projects.

Instructor and Program Lead | Bonyan Academy Aug 2025 - Apr 2026

- Delivered AI/programming instruction and coordinated instructors.

Full-Stack Development Intern | NTG Clarity Jul 2025 - Sep 2025

- Built a Spring Boot and Angular e-learning application with REST API integration.

Selected engineering projects

n8n Hackathon Cairo | 1st Place | Nov 2026

- Co-built an AI social media marketer in a five-person team in four hours: n8n research and writing agents, branded image generation, and human approval gates.

Lead Intelligence Pipeline | Tested reference **|** Code

- Engineered n8n workflows connecting lead intake to Odoo CRM, with deterministic scoring, separate LLM classification and workload-based routing.
- Implemented retries, a claim ledger, dead-letter replay and service-level (SLA) escalation to handle duplicate inputs and failed CRM writes.
- Passed 194 local assertions covering scoring, intake and error handling (Oct 2026). The deployed reference workflows are currently inactive.

Transcript Translator | Paid client delivery **|** Code

- Delivered an n8n pipeline for a paying client to translate DOCX transcripts into Arabic, accepting files through Telegram and returning a Google Drive link.
- Implemented chunked translation, timestamp preservation and ordered reassembly of the translated transcript.

AI Operating Systems | Two client implementations + personal

- Implemented tailored AI Operating Systems for two real-estate client organizations, alongside my own business AIOS, connecting business context, integrations, workflows and reusable agent capabilities.
- Built an original Egyptian-Arabic AIOS for one client and adapted integrations around another client's existing systems, with human approvals, documented operating rules and verifiable delivery boundaries.

LeadSync AI | Personal full-stack demo **|** Code

- Built Arabic Telegram lead intake, qualification and follow-up using Node.js, TypeScript and OpenRouter, with a Google Sheets CRM and manager dashboard.
- Implemented a conversation state machine and structured lead extraction for a fictional clinic scenario.

Inbox | Tested reference with simulated messaging

- Built a FastAPI, PostgreSQL and React shared inbox with qualification, transactional ownership claims, reassignment and audit flows.
- Validated the build with 261 backend tests and 48/48 smoke checks (Aug 2026), using a simulated messaging provider and PostgreSQL concurrency tests.

Education

BSc in Artificial Intelligence and Data Science | Expected 2027

Zewail City of Science, Technology and Innovation, Egypt

Graduation project: chest X-ray medical AI application integrating six deep-learning models, FastAPI, Angular and MongoDB, with explainability and hyperparameter optimization.

Community

Speaker, n8n Cairo Meetup #2 | Jul 2026. Presented Sawt, an Egyptian-Arabic voice-to-action showcase with tool integrations and human confirmation gates. 1st-place team winner, n8n Hackathon Cairo | Nov 2026.

Languages

Arabic | English